# HS Data Science 26

## Session 04: Scientific Working III — Sciometrics, Research Culture & Organisation

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# Session Overview

## Sciometrics & Research Quality

- measuring science: h-index, impact factor
- ranking venues: CORE, Scimago, DBLP
- predatory publishers and how to spot them
- open access and reproducibility

## Research Culture

- the replication crisis and what it means for ML
- open science principles
- research ethics and AI-specific concerns

## Organising Yourself

- Getting Things Done (GTD) for researchers
- weekly review as a habit
- tools: Zotero, Obsidian, Git research diary
- time-blocking and deep work

> Good science requires two things in equal measure: rigorous methods **and** sustainable work habits. Neither alone is enough.

# Measuring Science — Sciometrics

**Guiding Question:** How is scientific impact measured — and why does it matter which venue you publish in?

# The h-index

*Source: Hirsch 2005; Garfield 1955*

## Definition

A researcher has h-index **h** if exactly **h** of their papers have been cited at least **h** times (Hirsch 2005).

## Example

| Paper | Citations |
|---|---|
| P1 | 42 |
| P2 | 17 |
| P3 | 9 |
| P4 | 6 |
| P5 | 2 |

## Strengths

- combines productivity and impact in one number
- robust to one-off highly-cited outliers
- widely used by hiring committees and funding agencies

## Limitations

- favours quantity over quality of individual papers
- heavily field-dependent (ML h-indices are high; humanities are low)
- accumulates over time — disadvantages early-career researchers
- does not account for author order or self-citations

h-index = **4** (P1–P4 each have ≥ 4 cites)

**Where to check**

Google Scholar profile · Semantic Scholar · Publish or Perish (free)

Never compare h-indices across disciplines. An h-index of 15 means very different things in physics vs. computer science vs. literary studies.

# Impact Factor & Journal Metrics

**Impact Factor (IF)**

Average citations per paper published in a journal over the last 2 years.

$$IF = \frac{\text{citations to papers from years } t-1, t-2}{\text{papers published in years } t-1, t-2}$$

Published annually by Clarivate (Journal Citation Reports).

**CiteScore (Scopus)**

Similar idea, but uses a 4-year window — somewhat more stable.

**SJR (Scimago)**

Weighted by prestige of the citing journal — analogous to PageRank for journals.

**What it tells you**

- a rough signal of journal prestige within a field
- useful when selecting where to submit

**What it does NOT tell you**

- quality of any individual paper
- relevance to your specific research question
- reproducibility or rigour of published work

**For Computer Science**

Conference proceedings often matter more than journals. Many top venues (NeurIPS, ICML, ACL, WWW) have no impact factor — rank instead by CORE ranking or community acceptance.

> Impact Factor is a journal-level metric. It says nothing about whether a specific

# Ranking Research Venues — CORE

*Source: Research & Australasia 2023; Beall 2012*

## CORE Conference Ranking

The Computing Research and Education Association (CORE) ranks CS conferences:

| Rank | Meaning |
|---|---|
| **A*** | flagship, top tier worldwide |
| **A** | excellent, highly selective |
| **B** | good, respected |
| **C** | peer-reviewed, limited impact |
| Unranked | no quality signal |

**Check at:** portal.core.edu.au/conf-ranks

## How to use it

When reading a paper: check venue rank to calibrate how much trust to extend before diving in.

When submitting: target A*/A venues even if rejection is likely — reviewers' feedback is educational and forces quality.

## Warning: Predatory Venues

Signs of a predatory conference or journal:

- unsolicited invitation email promising fast review
- vague scope ("multidisciplinary", "all sciences")
- no indexing in DBLP, ACM DL, IEEE Xplore, Scopus

**For journals:** Scimago (SJR) Q1–Q4 quartiles per discipline

- excessive article processing charges with no clear affiliation
- check at: **beallslist.net**

> Publishing in a predatory venue harms your reputation and wastes time. Always verify before submitting.

# Open Science & Reproducibility

## The Replication Crisis

Many published results — particularly in psychology and medicine — fail to replicate when experiments are repeated. ML is not immune:

- code not released → results unverifiable
- hyperparameter tuning hidden → baselines cherry-picked
- dataset leakage → inflated metrics
- random seeds not reported → variance hidden

## Open Science Principles (UNESCO 2021)

1. Open data — share datasets where ethically possible
2. Open code — publish reproducible implementations
3. Open methods — describe setup completely

## Practical Steps for AI/ML Papers

- release code on GitHub with a clear README
- report all hyperparameters and hardware specs
- use fixed random seeds, report variance across multiple runs
- publish to arXiv pre-print before or alongside journal submission
- use **Papers With Code** to link results to implementations
- use model cards and datasheets for datasets (Mitchell et al., 2019)

## Reproducibility Checklists

NeurIPS and ICML now require reproducibility statements. Read them before writing — they are a free checklist of what reviewers will check.

4. Open peer review — transparency in the review process

Open, reproducible science is increasingly a requirement, not a bonus. Build the habit early.

# Research Culture & Ethics

# Research Ethics in AI

## Fundamental Principles

- **Honesty:** report results as they are, including negative results
- **Transparency:** describe methods fully enough for others to replicate
- **Accountability:** acknowledge limitations and failure modes
- **Non-maleficence:** consider harms of the research and its applications

## Specific AI Concerns

- bias in training data propagates to model outputs
- benchmark saturation creates false progress signals
- dual-use: the same model can help and harm
- environmental cost: large model training emits significant $CO_2$

## Academic Integrity

**Plagiarism:** using others' text or ideas without attribution — zero tolerance, automated detection (iThenticate, Turnitin).

**Fabrication / falsification:** inventing or manipulating data — career-ending, potentially criminal.

**Authorship:** every named author must have made a substantial intellectual contribution. Gift authorship and ghost authorship are both violations.

**AI-assisted writing:** LLMs may be used as a writing aid, but you are responsible for every claim. Disclose use per venue policy.

When in doubt: cite, disclose, and describe. The cost of over-attribution is zero; the cost of under-attribution can end a career.

# Organising Yourself — GTD for Researchers

**Guiding Question:** How do you manage a research project that has no clear "done" and spans months?

# Getting Things Done (GTD) — Core Workflow

*Source: Allen 2001*

## The Five Steps

1. **Capture** — collect every commitment, idea, task, and worry into a trusted inbox (notebook, phone, email folder)
2. **Clarify** — process each item: is it actionable? what is the next physical action?
3. **Organise** — put it in the right bucket: next-actions list, project list, waiting-for, someday/maybe, calendar
4. **Reflect** — weekly review: scan all lists, empty inbox, check project progress
5. **Engage** — choose what to work on based on context, energy, and priority

## The Research-Specific Buckets

| Bucket | Examples |
|---|---|
| Next Actions | "read Smith 2024 abstract", "run ablation on seed 42" |
| Projects | "seminar paper on RAG", "reproduce BEIR baseline" |
| Waiting For | "supervisor feedback on draft §3", "GPU cluster queue" |
| Someday/Maybe | "explore diffusion models for tabular data" |
| Reference | Zotero library, Obsidian notes, Git repo |



> Your brain is for thinking, not for storing. Capture everything externally and trust your system.

# The Weekly Review

**Why it matters:** Research has no natural stopping point. Without a regular review, tasks accumulate invisibly and projects stall.

### Weekly Review Checklist

- ▢ Process and empty inbox (email, notes, paper pile)
- ▢ Review last week's calendar — anything that needs follow-up?
- ▢ Check next-actions lists — mark done, identify stuck items
- ▢ Review all active projects — does each have a clear next action?
- ▢ Review waiting-for list — send follow-up nudges if overdue
- ▢ Review someday/maybe — promote anything now relevant

### Time and place

- 30–45 minutes, same time every week
- Friday afternoon is ideal (fresh week starts clean on Monday)
- physical separation helps: close all tabs first

### Tools that support this

| Purpose | Tool |
|---|---|
| Task management | Obsidian Tasks, Notion, plain text |
| Literature | Zotero (with Obsidian plugin) |
| Code & experiments | Git (commit = natural checkpoint) |

- ☐ Check upcoming calendar — prepare for commitments
- ☐ Identify the one most important thing for next week

| Purpose | Tool |
| --- | --- |
| Notes | Obsidian, Logseq, Roam |
| Calendar | Google Calendar (block deep work) |

> The weekly review is a habit, not a method. It takes four weeks to feel natural. Do not skip it.

# Deep Work and Research Time

*Source: Newport 2016*

## Cal Newport's Deep Work Framework

- **Deep work:** cognitively demanding tasks that push the limits of ability — writing, proving, designing experiments
- **Shallow work:** logistical tasks that can be done while distracted — email, administrative forms, formatting references

Research value comes almost exclusively from deep work.

### Key principles

1. Schedule deep work blocks (2–4 hours minimum) in the calendar
2. Eliminate distractions during deep work: phone off, notifications off

## Research Rhythms

| Block | Example |
|---|---|
| Morning deep work | writing, analysis, proofs |
| Mid-day shallow | email, Slack, administrative |
| Afternoon deep work | coding, experiments |
| Evening | reading papers (lighter cognitive load) |

## Protecting Deep Work

- batch email to two fixed times per day
- say no to low-value meetings

3. Measure deep work hours per week as a leading indicator
4. Rest deeply between blocks — shallow work in between is fine

- use "office hours" concept for collaboration requests
- log deep work hours (even a tally mark) — what gets measured improves

> Most researchers get 1–2 hours of genuine deep work per day. Aim for 4–6. That gap is where careers diverge.

# Research Tools Stack

## Literature Management — Zotero

- free, open-source reference manager
- browser plugin captures papers in one click
- syncs PDF highlights and annotations
- exports BibTeX directly for LaTeX
- Zotero Connector + Better BibTeX = seamless workflow

## Note-Taking — Obsidian

- plain Markdown files, no lock-in
- bidirectional links: paper notes link to concept notes link to project notes
- Zotero plugin: pulls metadata and annotation highlights automatically
- graph view reveals connection patterns in your reading

## The literature note template



## Code & Experiments — Git as Research Diary

Every commit message is a lab notebook entry:

```
feat: add BM25 baseline on MSMARCO dev split

Scores: MRR@10 = 0.187, Recall@100 = 0.857
Same as reported in Thakur et al. 2021 Table 3.
Confirms our reimplementation is correct.
Next: add dense retrieval (DPR).
```

## ML Project Structure

```
project/
├── data/           # raw (gitignored), processed
├── notebooks/      # exploration only, not production
├── src/            # importable modules
├── experiments/    # one folder per experiment run
│   └── 2026-05-12-bm25-baseline/
│       ├── config.yaml
│       ├── results.json
│       └── README.md
├── tests/
└── README.md
```

```
## Citation
Author (year). Title. Venue.

## Core Claim
One sentence.

## Evidence
How did they test it?

## Limitations & Gap
What does this leave open?

## Connection to My Work
```

Your future self is a stranger. Write notes and commit messages for them.

# Summary

## Sciometrics

- h-index = personal impact indicator (field-dependent!)
- Impact Factor = journal-level metric (not paper-level)
- CORE ranking = venue quality for CS conferences
- Predatory venues: always verify before submitting
- Open science: share code, data, methods

## Research Ethics

- honesty, transparency, accountability
- reproducibility is a professional obligation
- AI-specific: bias, dual-use, environmental cost
- plagiarism, fabrication, authorship norms

## Organisation

- GTD: Capture → Clarify → Organise → Reflect → Engage
- Weekly review: 30 min every Friday, non-negotiable
- Deep work: 4–6 hours/day is the goal
- Tools: Zotero + Obsidian + Git = complete research stack

**The researcher's compact:** Measure your impact without being ruled by it. Organise your work so that deep thinking gets protected time. Open your results so others can build on them.

# Image Credits & References


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